

Processing and interpretation of neuroscience data:

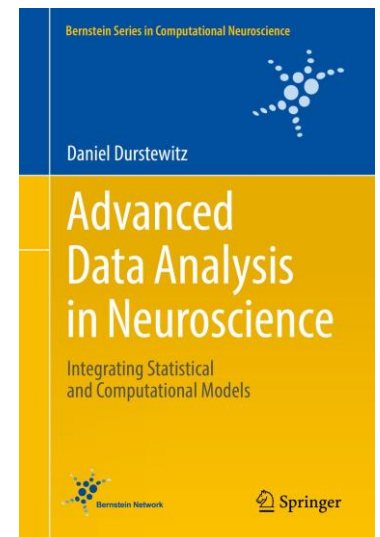
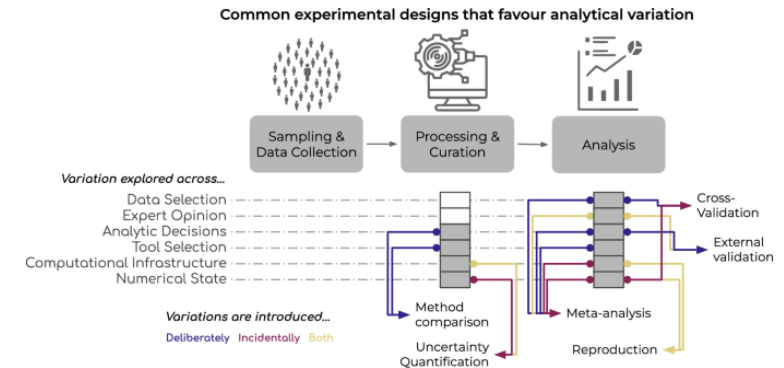
Module 1 – Data overview and single cell sequencing

Dr. Kaja Moczulska & Dr. Łukasz Piszczek

Univ.-Prof. Dr. Wulf Haubensak

Department of Neuronal Cell Biology

Center For Brain Research (Zentrum für Hirnforschung)



Overview

Resources needed:

- Laptop
- Internet access
- Materials:
 - VM machine - Docker
 - <https://github.com/HugoMalagon/NeuroData>
[860.053-MUW](#)

07.10

- Introduction to R, basics
- Visual analytics

14.10

- dimensional reduction: PCA, UMAP
- Normalization/scaling
- clustering: k-means, knn

21.10

- Intro to Seurat
- Single cell RNA seq
- Dataset merging and preprocessing

04.11

- clustering in Seurat
- Finding cell types

11.11

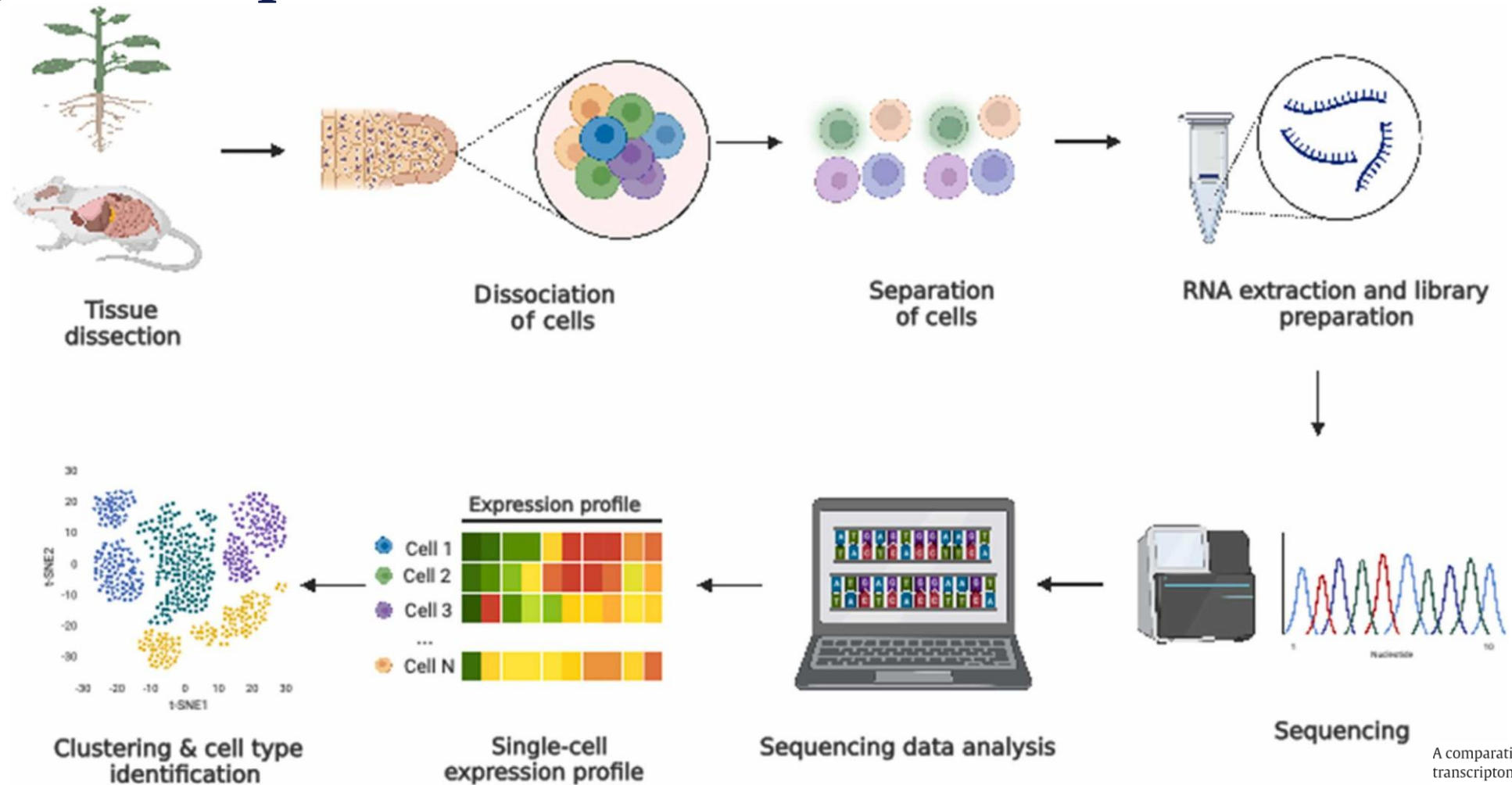
- Joining experiments
- DEG interpretation

„Exam“: Assessment via email after each class

Day 5

WT vs KO DEG analysis

Single cell experiments workflow



A comparative analysis of single-cell transcriptomic technologies in plants and animals

Vamsidhar Reddy Netti^{1,2}, Harshraj Shinde^{1,2}, Gulshan Kumar^{1,2}, Ambika Dudhate^{1,2}, Jong Chan Hong^{1,2}, A. B. Uthas Sapanrao Kadam^{1,2}

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<https://doi.org/10.1016/j.cpb.2021.100289>

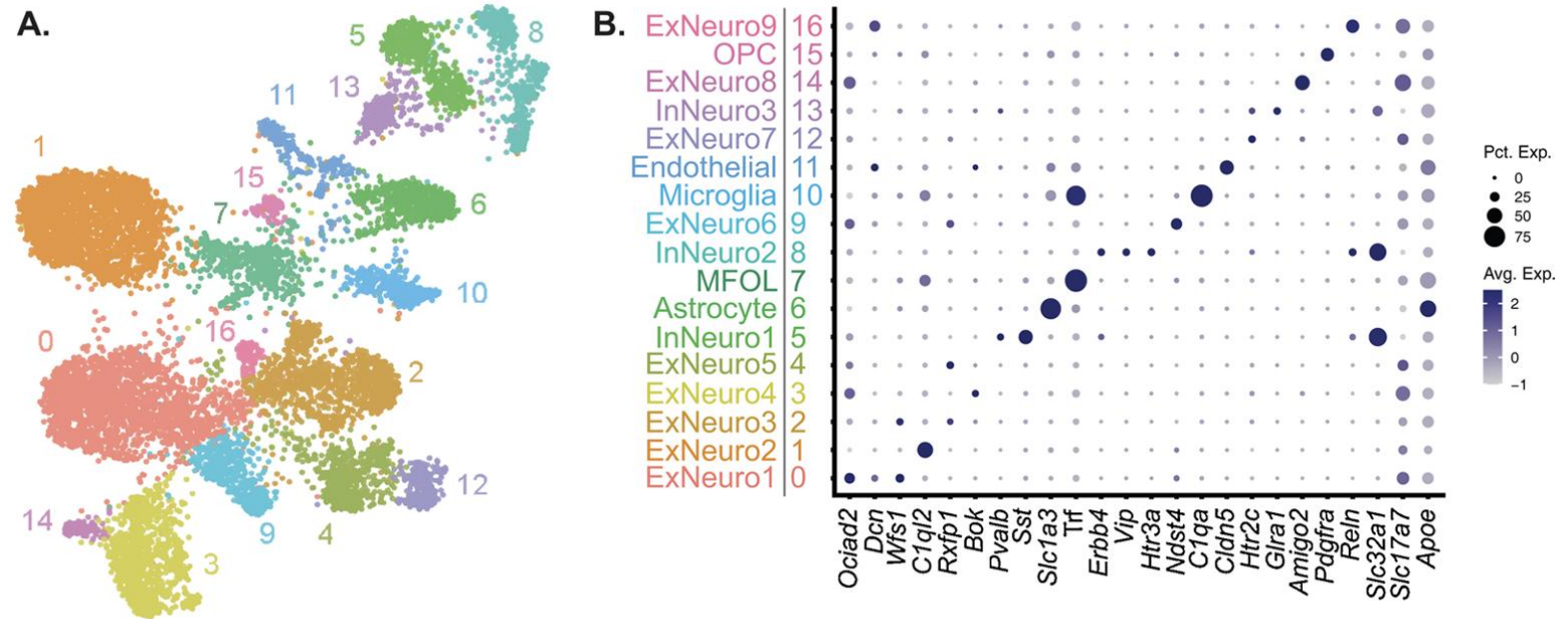
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Dataset we will analyse today

- Alzheimer's disease model: ApoE knock out
- Single nucleus data
- 7 pooled hippocampi samples from 9-month-old C57BL/6 J wild type (WT) mice (ID GSM5067107)
- 7 pooled hippocampi samples from 9-month-old *ApoE* knockout (EKO) mice (ID GSM5067109)

differences in specific cell types.

All analyses were performed on GEO dataset [GSE166261](#) [dataset] (Shi et al., 2021a, b) comprised of snRNAseq data generated by droplet-based Chromium Single Cell 3' Reagent Kit (10x Genomics) on 7 pooled hippocampi samples from 9-month-old C57BL/6 J wild type (WT) mice (Sample GSM5067107) and 7 pooled hippocampi samples from 9-month-old *ApoE* knockout (EKO) mice (Sample GSM5067109). The initial EKO mouse was generated using the 129P2/OlaHsd-derived E14Tg2a ES cell line (Piedrahita et al., 1992). The mice used to generate the samples for the snRNAseq data for this study were produced by backcrossing at least 10 generations to C57BL/6 J inbred mice.



Short Communication

Analysis of differential gene expression and transcript usage in hippocampus of *ApoE* null mutant mice: Implications for Alzheimer's disease

Andrew E. Weller*, Glenn A. Doyle, Benjamin C. Reiner, Richard C. Crist, Wade H. Berrettini

Center for Neurobiology and Behavior, Department of Psychiatry, Perelman School of Medicine, University of Pennsylvania, Philadelphia, PA, 19104, United States

General workflow

- Data import and Seurat object generation (already done)
- QC for mitochondrial genes, nr of genes and expression level
- Normalization: normalizing gene expression level for each cell
- Integration of data sets (AD and WT mice)
- Feature selection of variable genes
- Scaling: scaling expression for each gene to avoid overrepresentation of highly expressed genes
- Linear dimensional reduction: PCA
- Clustering: k-means for cluster selection
- Non-linear dimensional reduction: UMAP
- Feature selection and cluster annotation: looking at cluster specific markers
- (optional) Cell mapping, subclustering

Dataset integration

Why we should integrate the data?

- compare gene expression patterns between different conditions
- identify common and rare cell types
- build a large reference atlas
- ...

What the source of data variation?

- technical factors
 - assays
 - platform
 - Protocol...
- biological variation
 - distinct tissue sampling regions
 - different genotype
 - different genetic background
 - different age
 - different species
 - different pre-treatment...

Data integration methods

- Promote cell type identity over background identity
 - in similar datasets
 - in different datasets
- The datasets should contain analogous cell types
- Popular algorithms applied
 - principal component analysis (PCA)
 - singular value decomposition (SVD)
 - canonical correlation analysis (CCA)

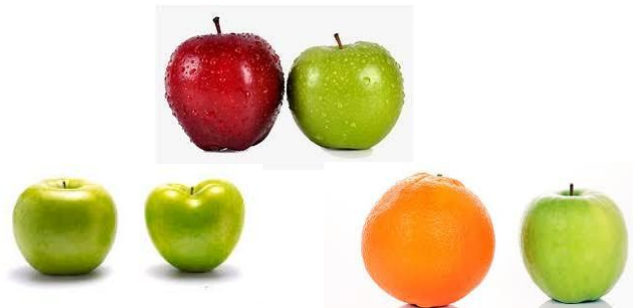
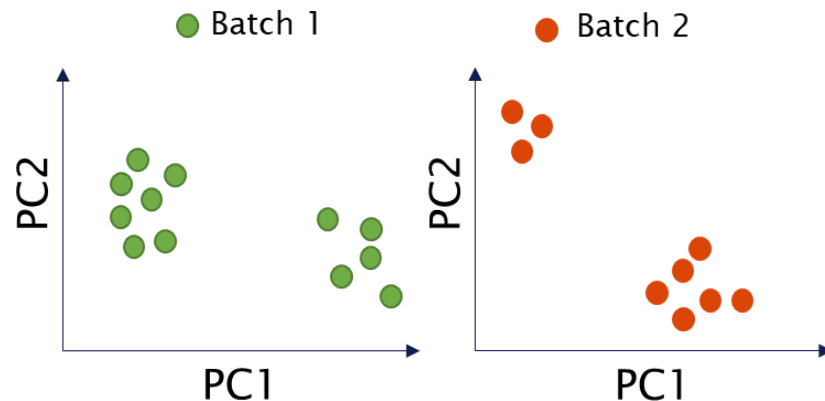


Table 6. Representative methods for single-cell and spatial transcriptomics integration.

Tool	Input Data Demonstrated	scRNA-seq Data Preprocessing	Methods/Algorithms	Application/Output
Mapping				
scVI	scRNA-seq <-> scRNA-seq	Raw count matrix	Probabilistic modeling, neural networks, variational inference	scRNA-seq cell level and gene level batch correction scRNA-seq mapping
scANVI	scRNA-seq <-> scRNA-seq	Raw count matrix for UMI counts, gene length normalized count for read counts	Probabilistic modeling, neural networks, variational inference	scRNA-seq cell level and gene level batch correction scRNA-seq mapping annotation of single cells from annotated reference cells
MNN/fastMNN	scRNA-seq <-> scRNA-seq	Normalized with the library size, log transformed	Randomized SVD, MNN, weighted average of correction vectors	scRNA-seq cell level and gene level batch correction scRNA-seq mapping
Scanorama	scRNA-seq <-> scRNA-seq	L2-normalized for each cell	Randomized SVD, MNN, weighted average of correction vectors	scRNA-seq cell level and gene level batch correction scRNA-seq mapping
Seurat V3	scRNA-seq <-> scRNA-seq scRNA-seq <-> HPRI scRNA-seq <-> CITE-seq scRNA-seq <-> scATAC-seq	Normalized with the library size, log transformed, gene scaled	CCA, MNN, anchor scoring and weighting	scRNA-seq cell level and gene level batch correction scRNA-seq mapping Multimodal data mapping
Harmony	scRNA-seq <-> scRNA-seq scRNA-seq <-> HPRI	Normalized with the library size, log transformed, gene scaled, PCs from PCA	Maximum batch diversity soft k-means clustering, linear mixture model correction	scRNA-seq cell level batch correction scRNA-seq mapping Multimodal data mapping
LIGER	scRNA-seq <-> scRNA-seq scRNA-seq <-> HPRI scRNA-seq <-> single cell DNA methylation	Normalized with the library size, gene scaled but not centered	Integrative non-negative matrix factorization, shared factor neighborhood clustering	scRNA-seq cell level batch correction scRNA-seq mapping Multimodal data mapping
SpaGE	scRNA-seq <-> HPRI	Normalized with the library size, log transformed, gene scaled	SVD on the cosine similarity matrix of PCs from each modality	Multimodal data mapping
gimVI	scRNA-seq <-> HPRI	Raw count matrix	Probabilistic modeling, neural networks, variational inference	Multimodal data mapping (for gene imputation)
Tangram	scRNA-seq <-> spatial barcoding and HPRI	Normalized with the library size	Direct minimization of a Kullback–Leibler divergence and cosine distances	Multimodal data mapping
Cobolt	Unimodal data sets and multimodal data set	Raw count matrix	Probabilistic modeling, neural networks, variational inference	Multimodal data mapping
MultiVI	Unimodal data sets and multimodal data set	Raw count matrix	Probabilistic modeling, neural networks, variational inference	Multimodal data mapping
Seurat V5	Unimodal data sets and multimodal data sets	Depends on the mapping method in the first mapping step	Dictionary learning, Laplacian eigen-decomposition, sketching	Multimodal data mapping
Deconvolution				
Cell2location	scRNA-seq <-> spatial barcoding data	Raw count matrix	Bayesian negative binomial models, approximate variational inference	Estimate the absolute cell type abundance for each spot of spatial data
RCTD	scRNA-seq <-> spatial barcoding data	Raw count matrix	Poisson–lognormal models	Estimate the proportion of cell types for each spot of spatial data
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DestVI	scRNA-seq <-> spatial barcoding data	Raw count matrix	Probabilistic modeling, negative binomial models, neural networks, variational inference	Estimate both the proportion of cell types and the variations within each cell type for each spot of spatial

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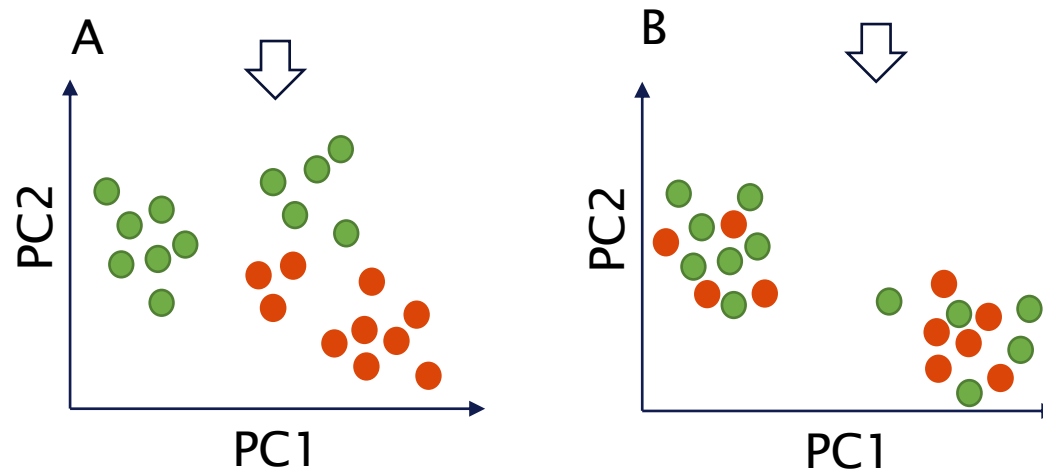
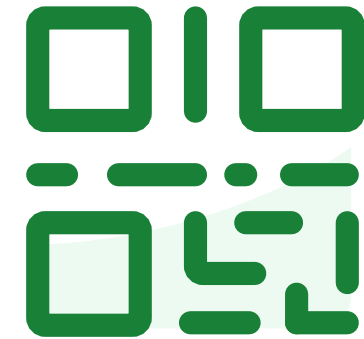
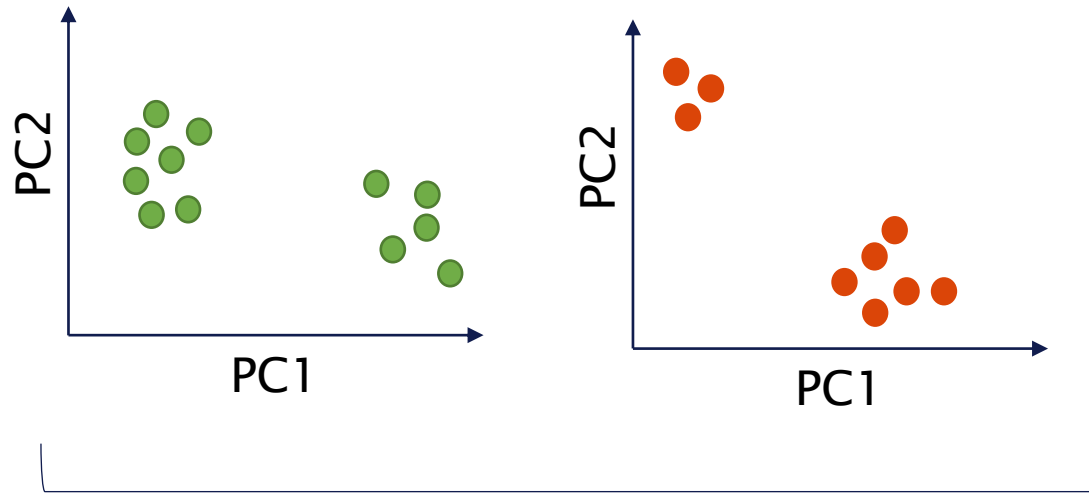
A Review of Single-Cell RNA-Seq Annotation, Integration, and Cell–Cell Communication

by Changde Cheng ^{1,†} , Wenan Chen ^{2,†} , Hongjian Jin ^{2,†}  and Xiang Chen ^{1,*} 

Comparing different data sets – data integration

● Batch 1

● Batch 2



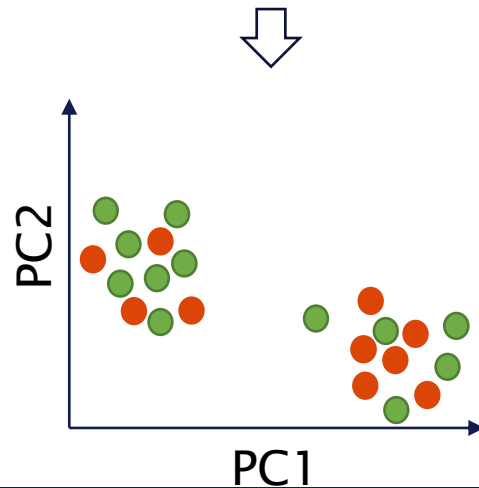
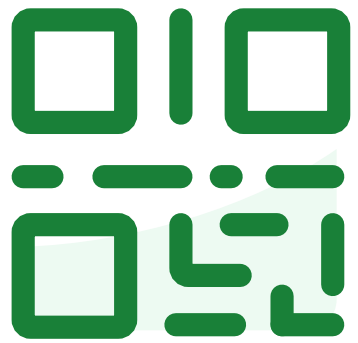
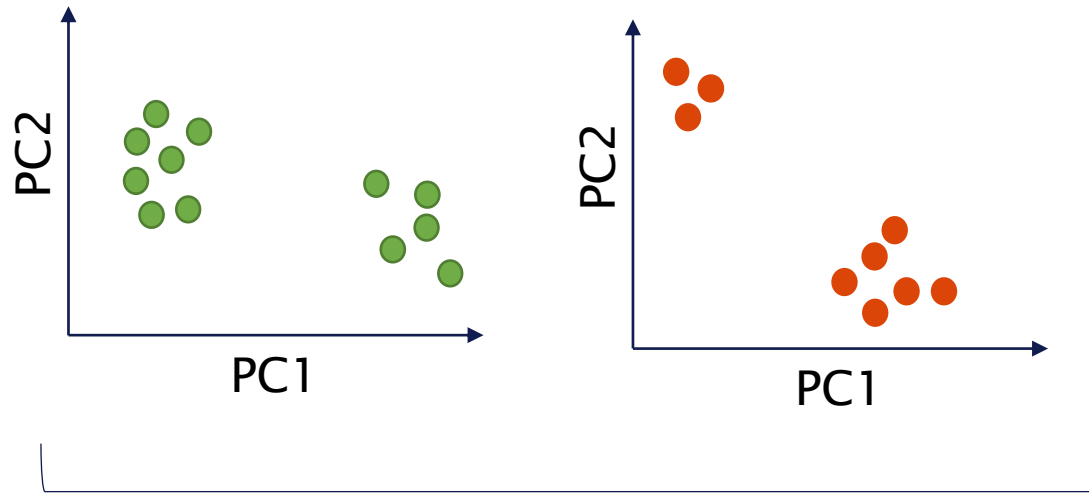


Which outcome do you prefer?

Comparing different data sets – data integration

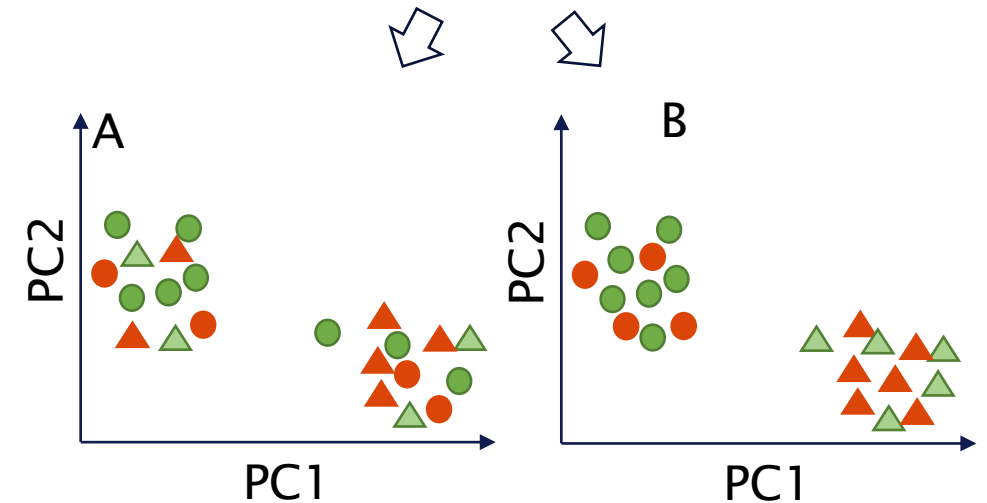
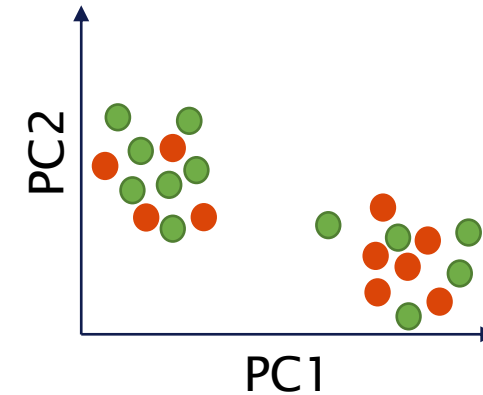
● Dataset 1

● Dataset 2



○ Cell type A

△ Cell type B





Which clusters represent better cell types?

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A Review of Single-Cell RNA-Seq Annotation, Integration, and Cell–Cell Communication

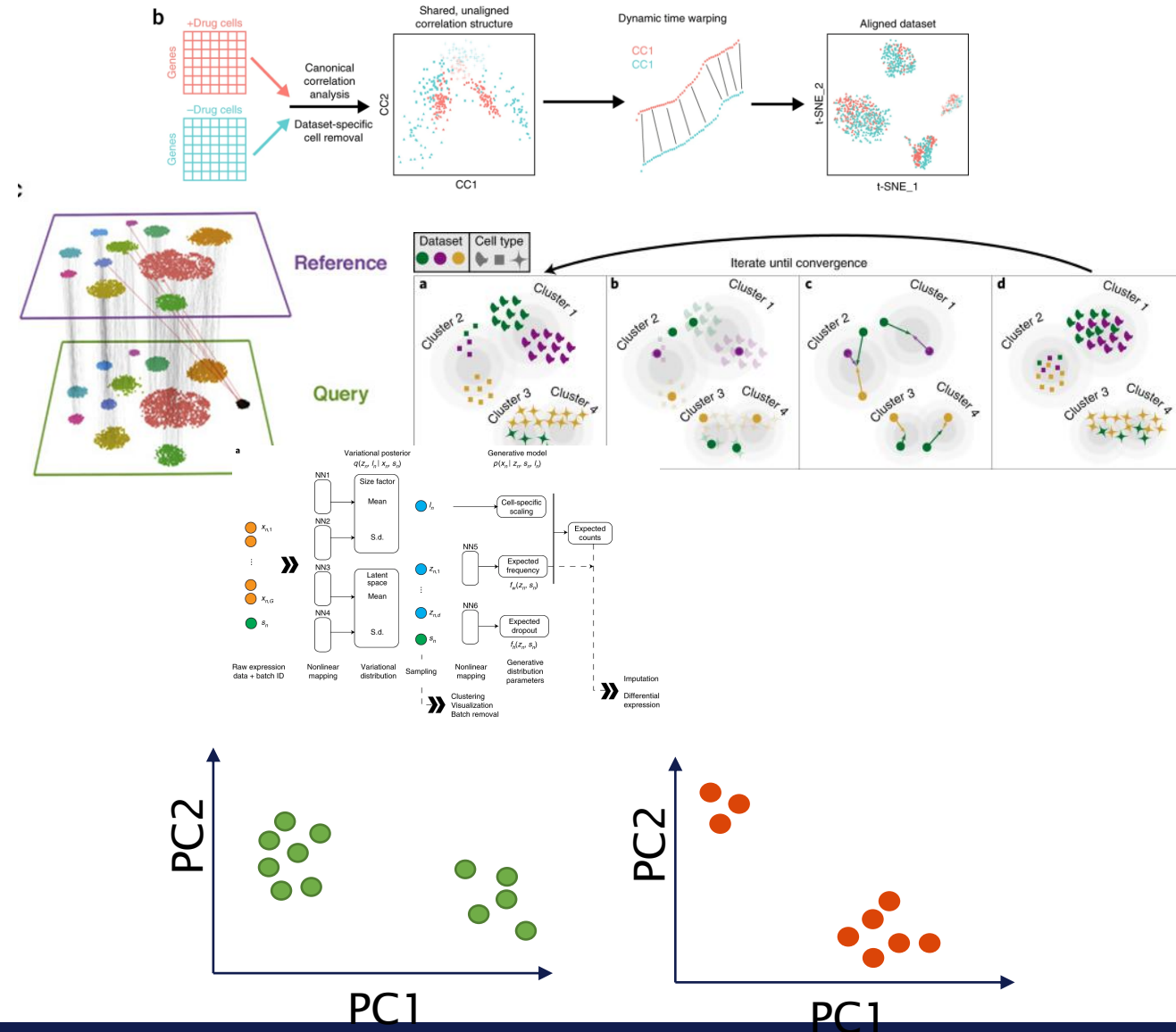
 by Changde Cheng ^{1,†} , Wenan Chen ^{2,†} , Hongjian Jin ^{2,†}  and Xiang Chen ^{1,*} 

Choice of methods to integrate datasets in Seurat

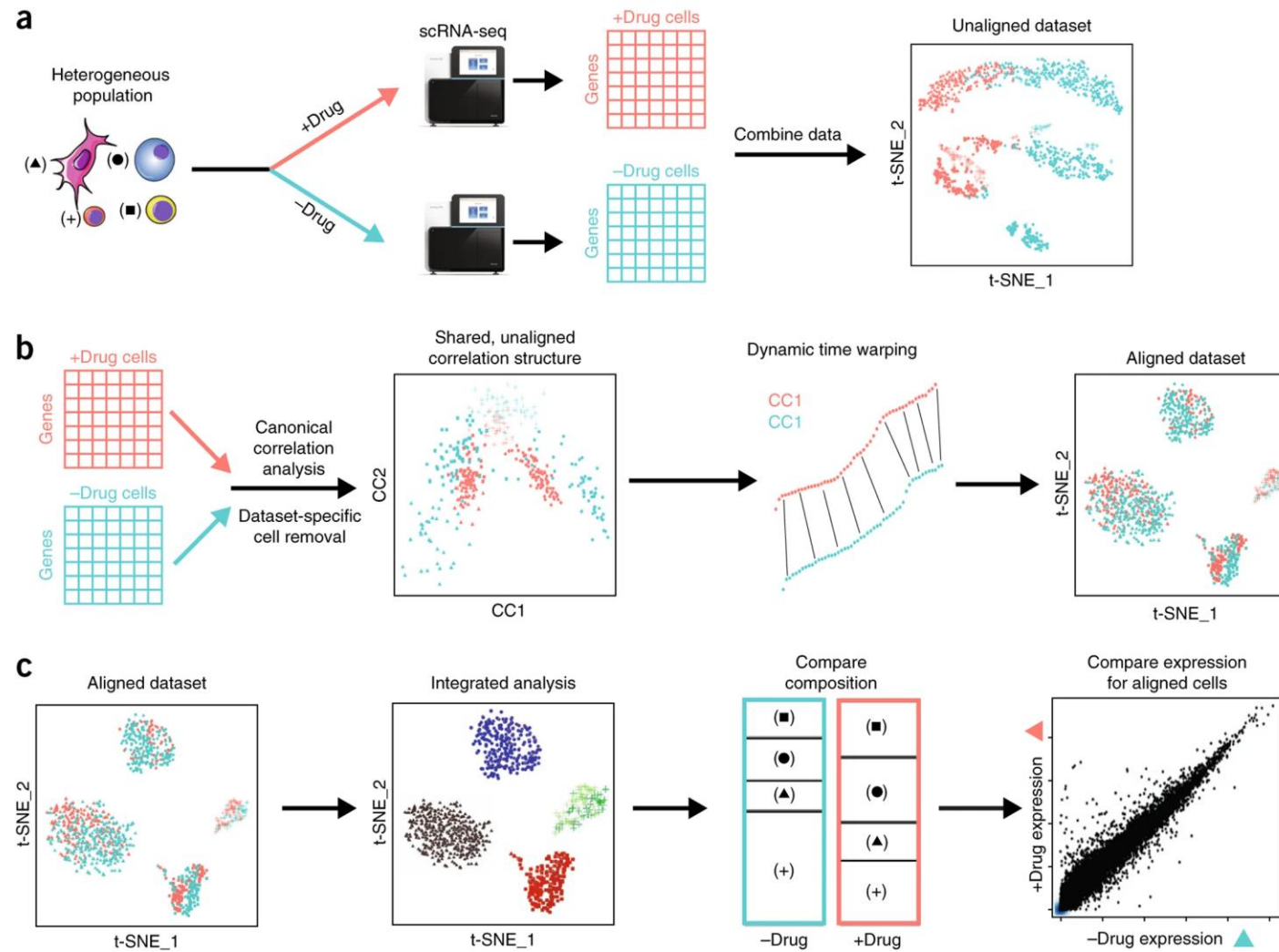
- RPCA – reciprocal PCA
- CCA - canonical correlation analysis
- Harmony – iterative clustering
- scVI – neural network model

What influences the choice of method

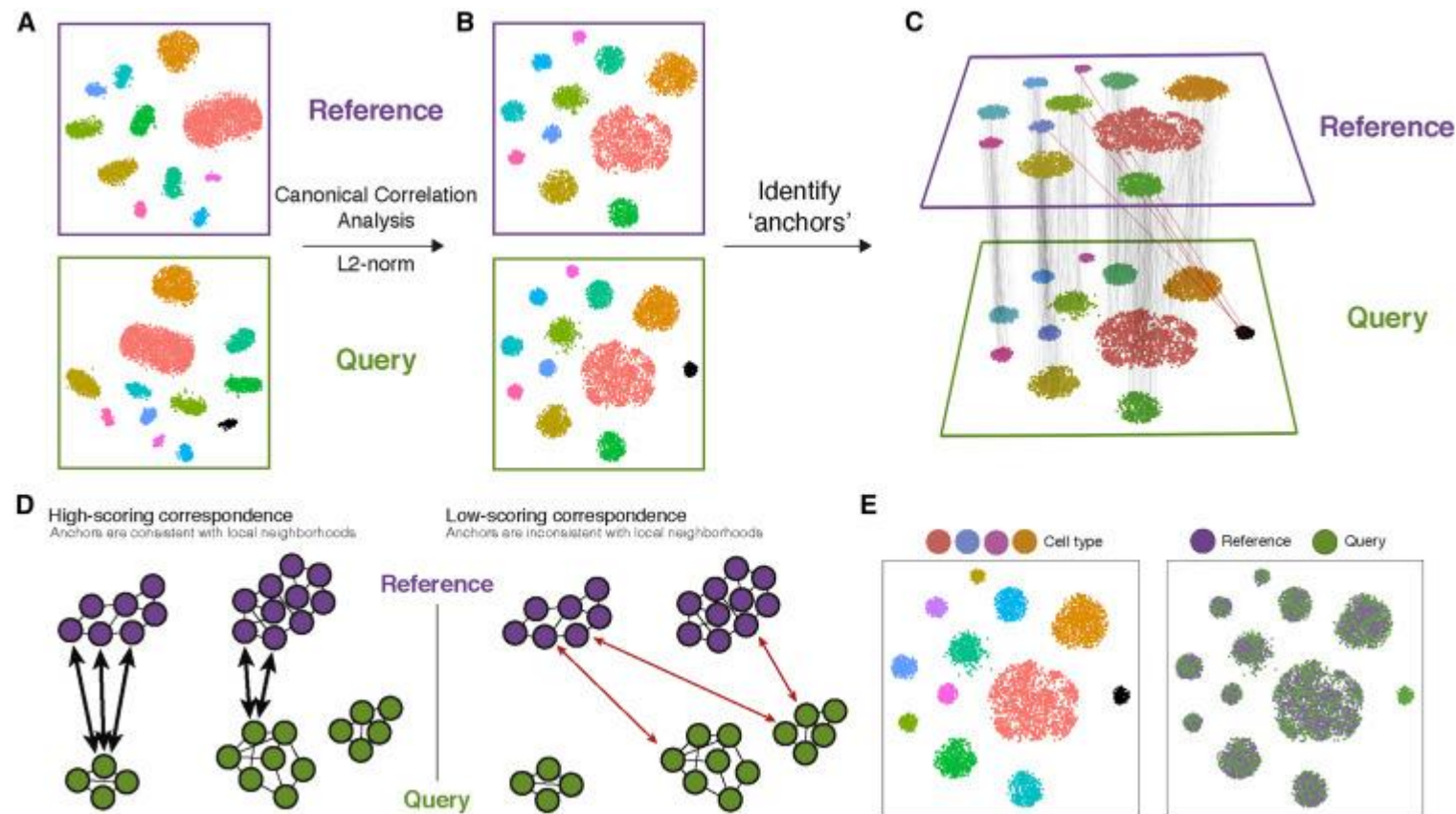
- How similar are 2 datasets
- How big is the data
- Scalability of method
- Size differences between the datasets
- Batch effect



CCA (canonical correlation analysis)



RPCA (reciprocal PCA)



Cell

Volume 177, Issue 7, 13 June 2019, Pages 1888–1902.e21

Resource

Comprehensive Integration of Single-Cell Data

Tim Stuart^{1,4}, Andrew Butler^{1,2,4}, Paul Hoffman¹, Christoph Hafemeister¹, Efthymia Papalexi^{1,2}, William M. Mauck III^{1,2}, Yuhao Hao^{1,2}, Marlon Stoeckius¹, Peter Smibert¹, Rahul Satija^{1,2,5}

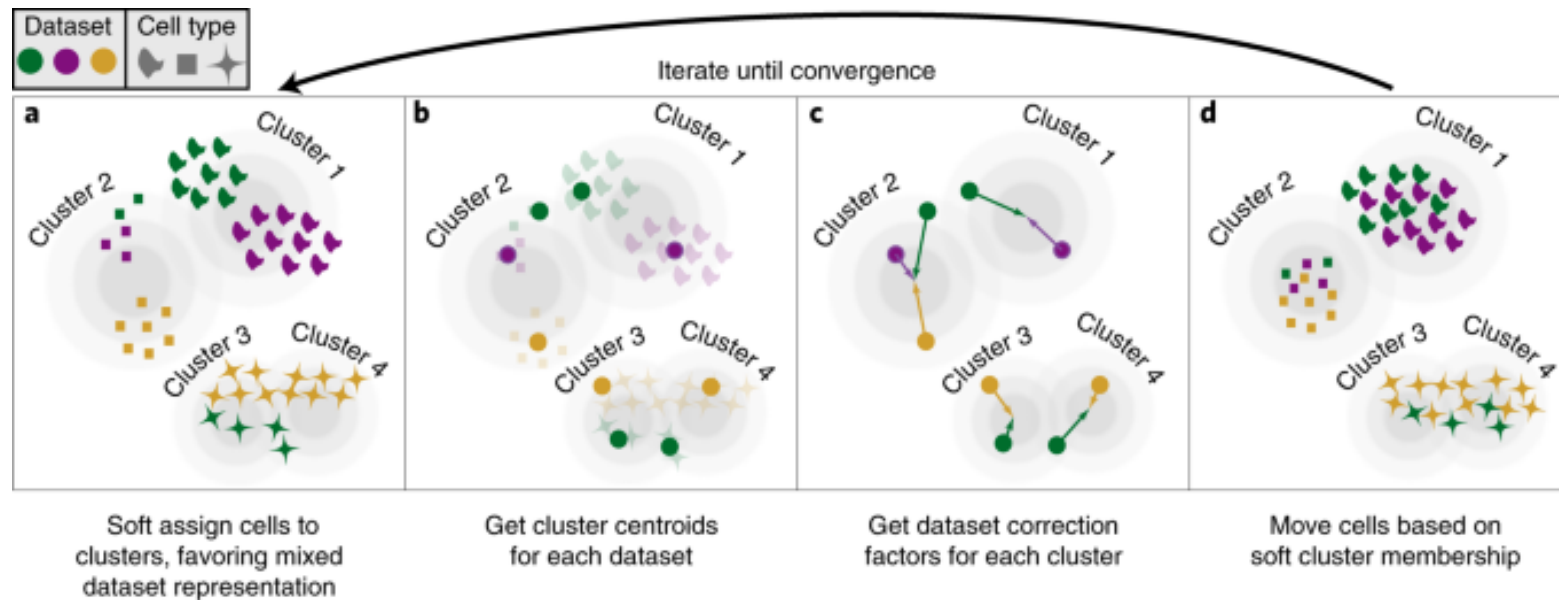
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<https://doi.org/10.1016/j.cell.2019.05.031>

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Harmony



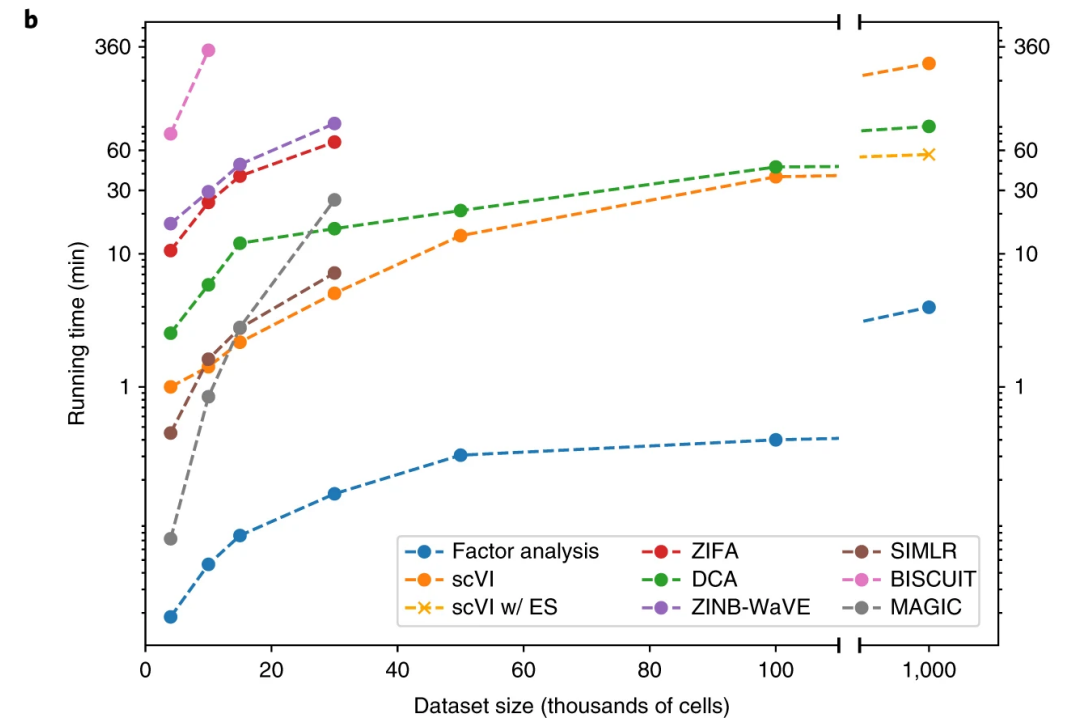
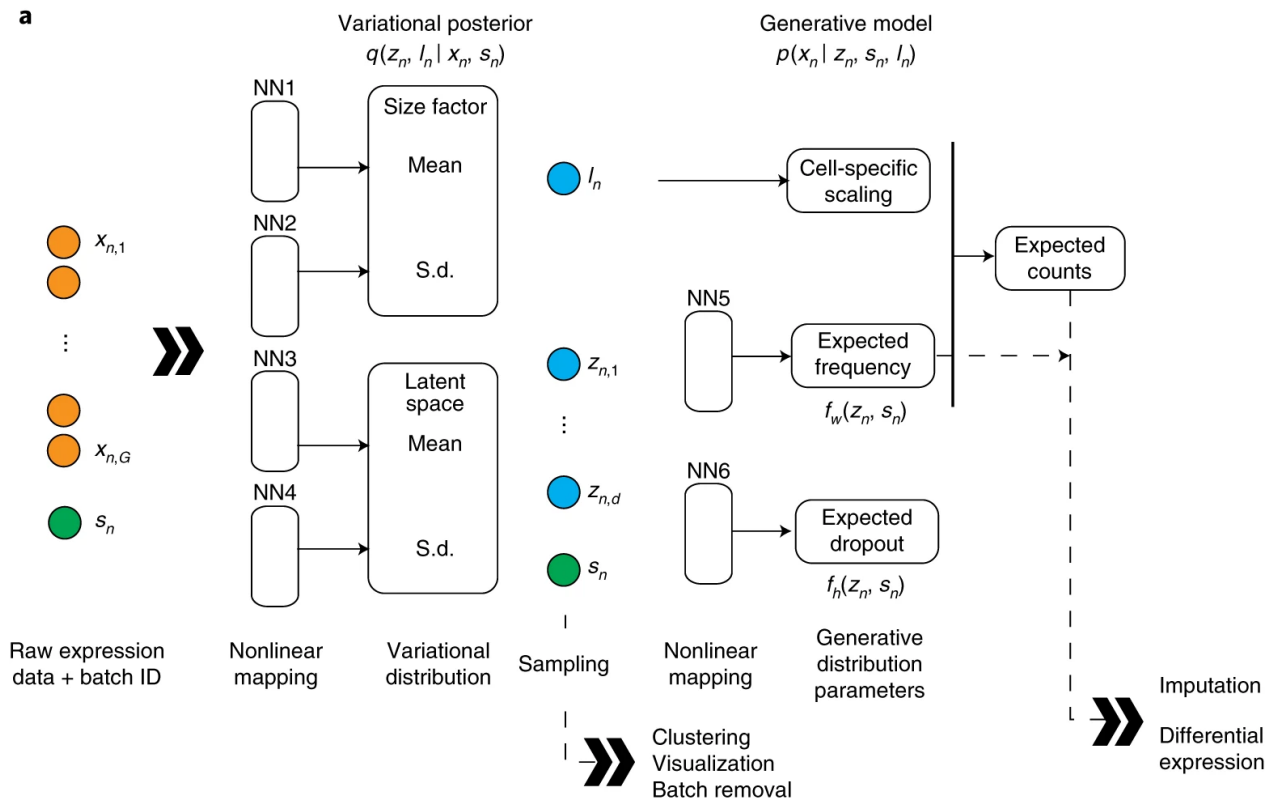
Article | Published: 18 November 2019

Fast, sensitive and accurate integration of single-cell data with Harmony

Ilya Korsunsky, Nghia Millard, Jean Fan, Kamil Slowikowski, Fan Zhang, Kevin Wei, Yuriy Baglaenko, Michael Brenner, Po-ru Loh & Soumya Raychaudhuri

Nature Methods **16**, 1289–1296 (2019) | [Cite this article](#)

scVI (single-cell variational inference) neural networks model



Article | Published: 30 November 2018

Deep generative modeling for single-cell transcriptomics

Romain Lopez, Jeffrey Regier, Michael B. Cole, Michael I. Jordan & Nir Yosef

Nature Methods 15, 1053–1058 (2018) | [Cite this article](#)

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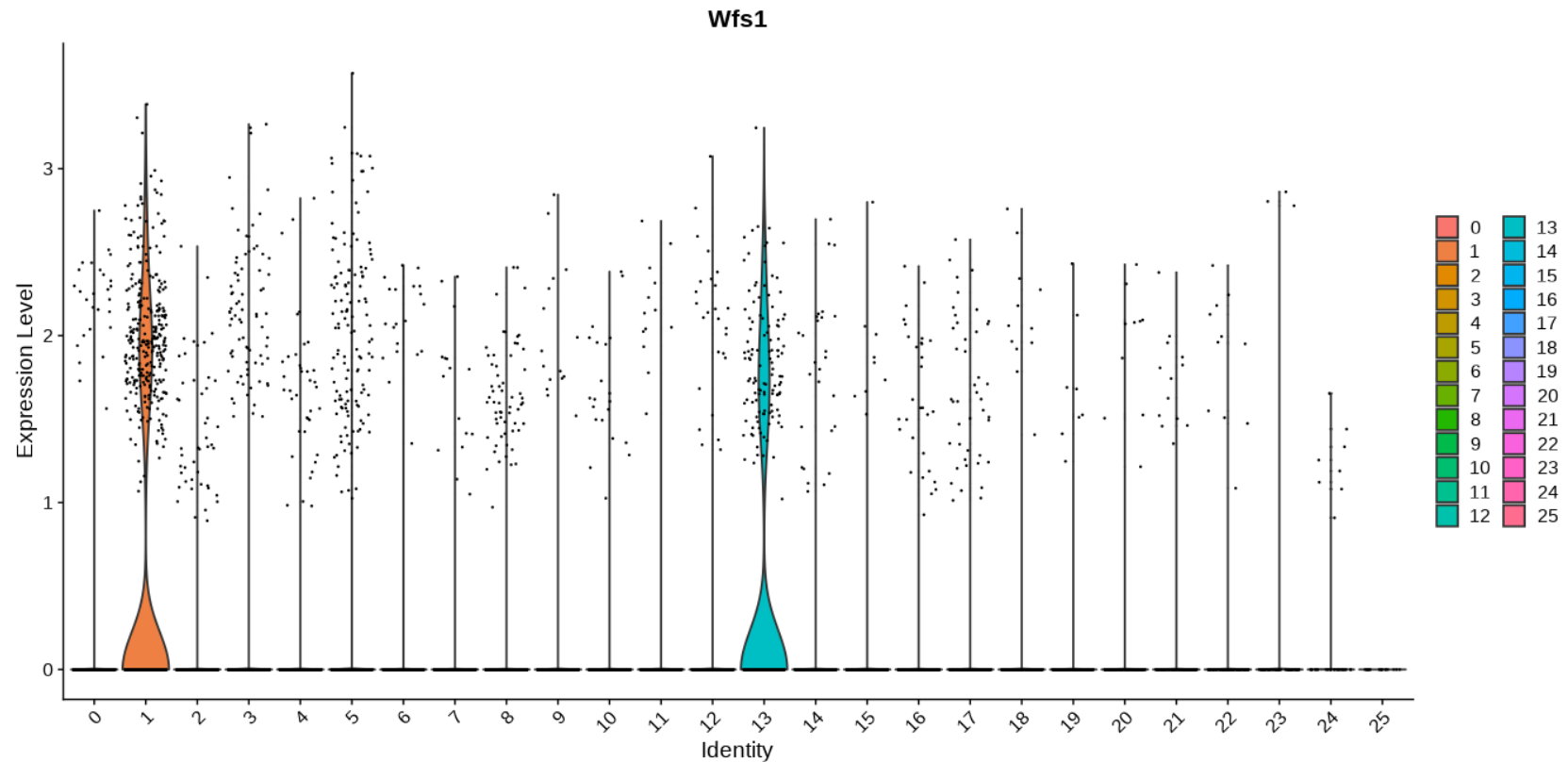
General workflow

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Part 8: Find markers

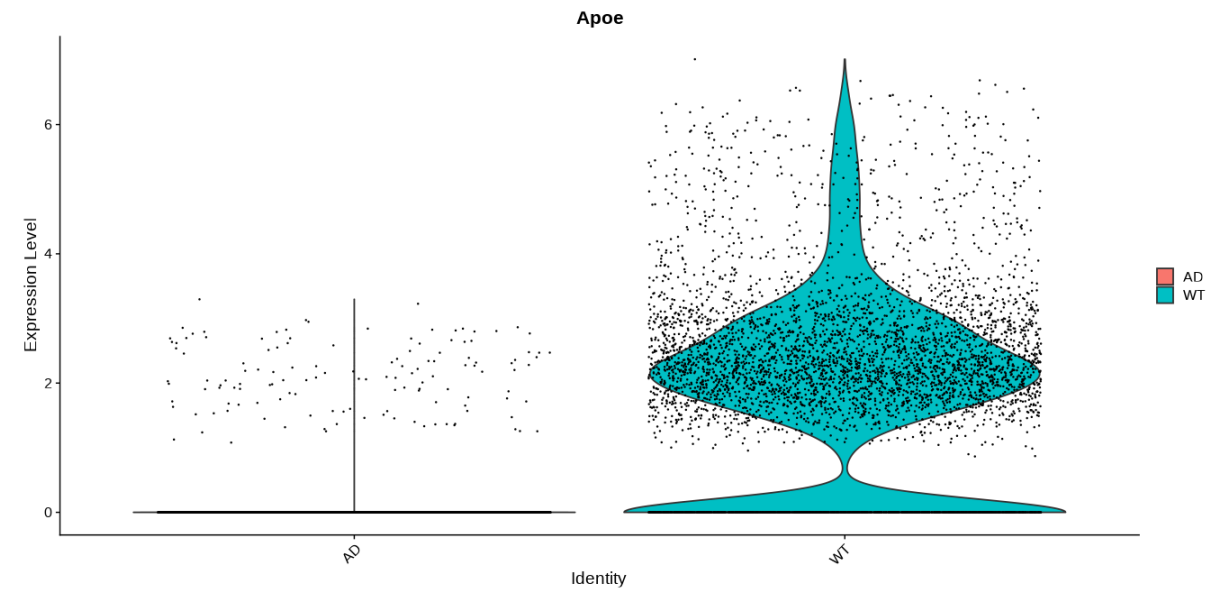
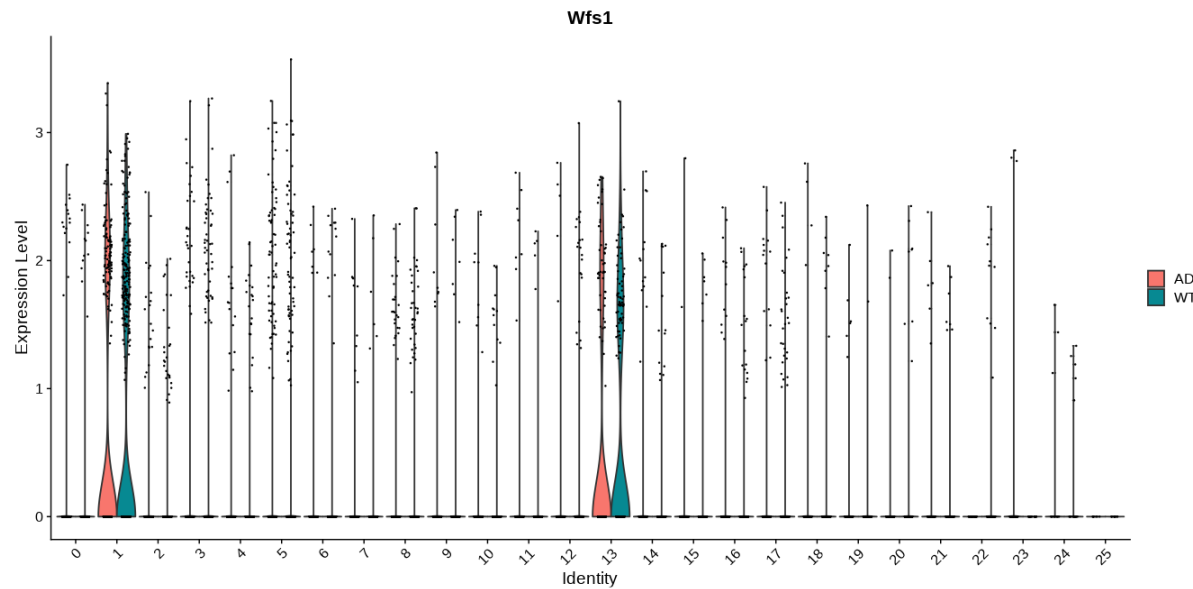
Differential gene expression

```
VlnPlot(obj, 'D1x2')
```



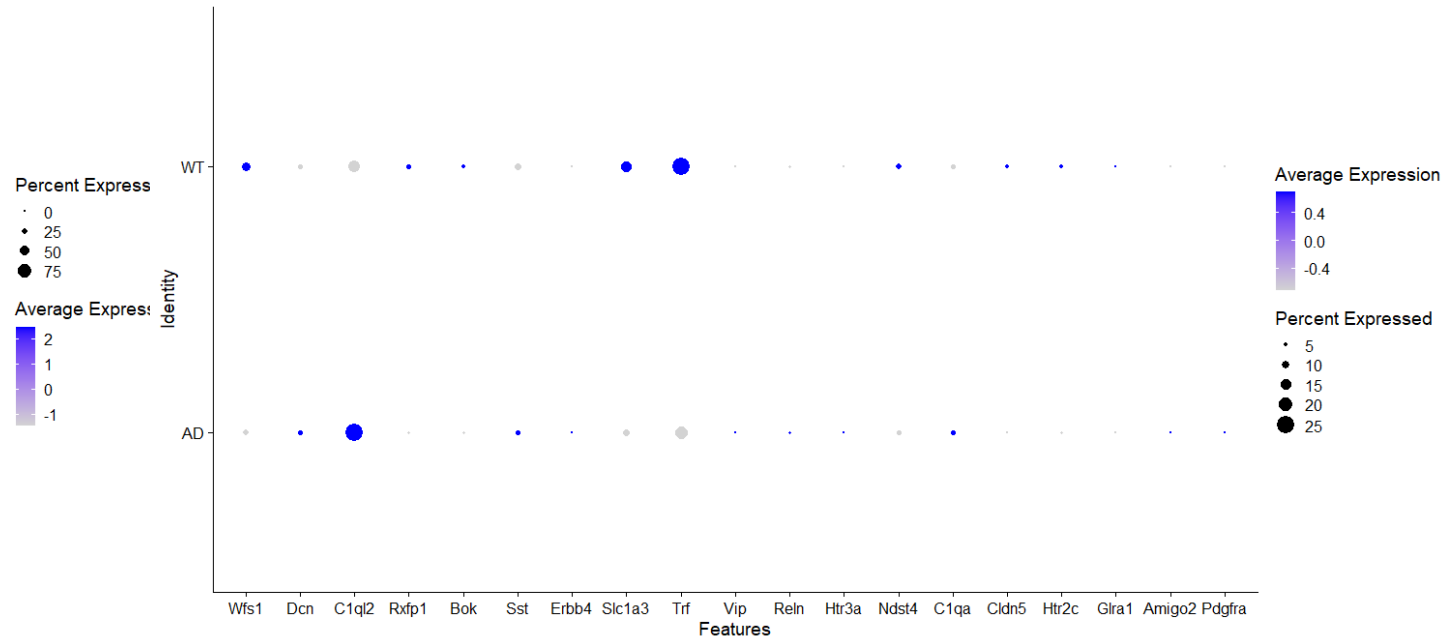
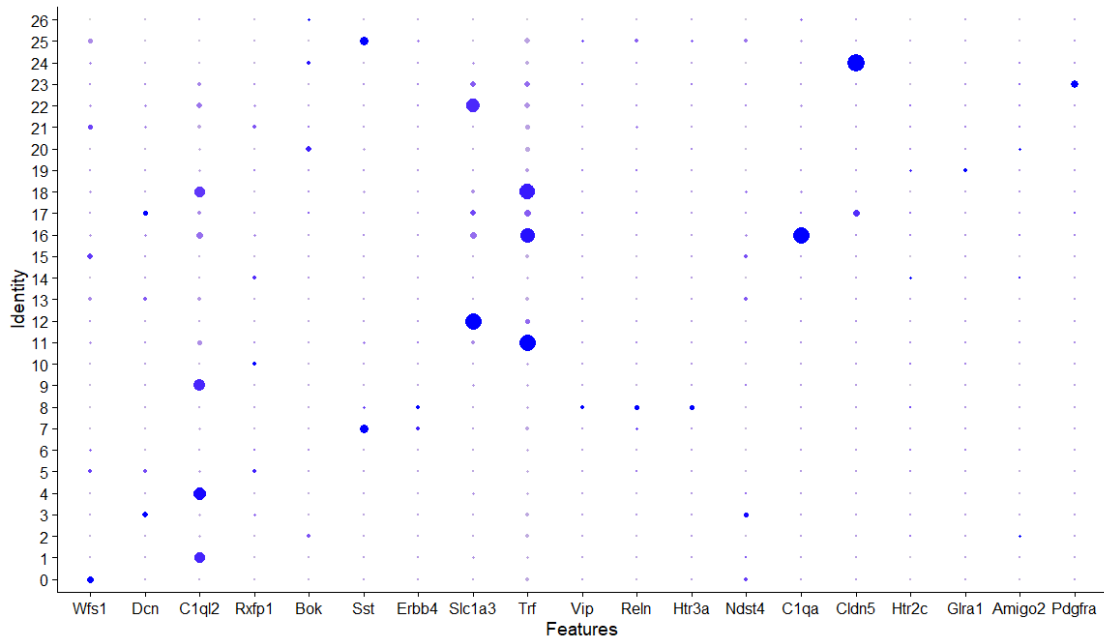
Part 8: Find markers

```
```{r, fig.height = 7, fig.width = 14}
VlnPlot(Seurat_AD_WT, 'Wfs1')
VlnPlot(Seurat_AD_WT, 'Wfs1', split.by = 'condition')
VlnPlot(Seurat_AD_WT, 'Apoe', group.by = 'condition') # check knockout
```
```



Part 8: Find markers

```
```{r, fig.height = 7, fig.width = 14}
DotPlot(Seurat_AD_WT, features= all_genes)
DotPlot(Seurat_AD_WT, features= all_genes, group.by = "condition")
```
```



Part 8: Find markers

```
```{r, fig.height = 7, fig.width = 14}
find all markers of cluster 5
Idents(Seurat_AD_WT) <- "seurat_clusters"
cluster13.markers <- FindMarkers(Seurat_AD_WT, ident.1 = "13", only.pos = TRUE)
head(cluster13.markers, n = 10)
```

## Markers

	p_val	avg_log2FC	pct.1	pct.2	p_val_adj
Cpne7	3.795059e-113	1.6148848	0.939	0.440	1.062541e-108
Pex5l	6.549430e-80	1.3841500	0.867	0.414	1.833709e-75
Man1a	1.168222e-55	1.7415038	0.408	0.123	3.270787e-51
Grin2b	6.088627e-55	0.6456170	1.000	0.913	1.704694e-50
Kcnn2	1.985708e-46	1.3211142	0.589	0.252	5.559586e-42
3110035E14Rik	1.610460e-42	1.1141428	0.614	0.279	4.508965e-38
Ntn	1.879055e-40	0.7167477	0.956	0.618	5.260977e-36
Chrd	8.107074e-40	1.0820214	0.683	0.354	2.269819e-35
Fam19a1	3.278474e-37	0.6712845	0.917	0.568	9.179072e-33
Fibcd1	4.322958e-37	1.6778961	0.267	0.076	1.210342e-32
Zeb2	6.237067e-33	0.7754559	0.853	0.579	1.746254e-28
Arhgef28	8.172101e-31	1.3445675	0.294	0.099	2.288025e-26
Mpped1	4.221407e-29	1.0897637	0.481	0.220	1.181910e-24
Brd9	7.888922e-28	0.7317513	0.836	0.619	2.208740e-23
Cadps2	6.051534e-27	1.3831535	0.256	0.086	1.694308e-22
Atp2b1	1.206030e-26	0.5222782	0.939	0.733	3.376643e-22
Grm5	2.330329e-26	0.5810710	0.939	0.721	6.524456e-22

- **avg\_logFC**: log fold-change of the average expression between the two groups. Positive values indicate that the gene is more highly expressed in the first group
- **pct.1**: The percentage of cells where the gene is detected in the first group
- **pct.2**: The percentage of cells where the gene is detected in the second group
- **p\_val\_adj**: Adjusted p-value, based on bonferroni correction using all genes in the dataset

# Part 8: Find markers - Finds markers that are conserved between the groups

```
Idents(Seurat_AD_WT) <- "seurat_clusters"
cluster13.conserved <- FindConservedMarkers(Seurat_AD_WT, ident.1 = "13", grouping.var = "condition", verbose = FALSE)
```

## Conserved Markers

	WT_p_val	WT_avg_log2FC	WT_pct.1	WT_pct.2	WT_p_val_adj	AD_p_val	AD_avg_log2FC	AD_pct.1	AD_pct.2	AD_p_val_adj	max_pval	minimum_p_val
Cpne7	1.896286e-53	1.5667888	0.921	0.475	5.309222e-49	3.859953e-62	1.6734406	0.959	0.406	1.080710e-57	1.896286e-53	7.719906e-62
Man1a	4.180960e-17	1.4020101	0.372	0.149	1.170585e-12	7.693521e-47	2.1012876	0.450	0.097	2.154032e-42	4.180960e-17	1.538704e-46
Pex5l	2.917237e-40	1.4236271	0.859	0.435	8.167680e-36	2.128116e-41	1.3607309	0.876	0.395	5.958298e-37	2.917237e-40	4.256231e-41
Grin2b	7.679557e-37	0.7202418	1.000	0.929	2.150122e-32	1.357831e-20	0.5502120	1.000	0.896	3.801656e-16	1.357831e-20	1.535911e-36
Kcnn2	1.643295e-19	1.3106818	0.571	0.288	4.600898e-15	6.643690e-30	1.3249140	0.609	0.216	1.860100e-25	1.643295e-19	1.328738e-29
Fam19a1	6.108650e-26	0.7810839	0.942	0.578	1.710300e-21	6.415784e-14	0.5597053	0.888	0.557	1.796291e-09	6.415784e-14	1.221730e-25
Arhgef28	8.345182e-10	1.1449230	0.262	0.114	2.336484e-05	1.180007e-25	1.5509784	0.331	0.085	3.303784e-21	8.345182e-10	2.360014e-25
Ntn1	6.855173e-18	0.6564655	0.953	0.702	1.919311e-13	2.629994e-25	0.7718903	0.959	0.536	7.363456e-21	6.855173e-18	5.259987e-25
3110035E14Rik	7.409298e-20	0.9887040	0.660	0.348	2.074455e-15	3.045912e-25	1.2567730	0.562	0.212	8.527944e-21	7.409298e-20	6.091824e-25
Chrd	1.333034e-18	1.0821957	0.660	0.375	3.732227e-14	4.238994e-23	1.0950006	0.710	0.334	1.186834e-18	1.333034e-18	8.477988e-23
Zeb2	4.148900e-21	0.8652456	0.853	0.613	1.161609e-16	1.146133e-13	0.6508298	0.852	0.546	3.208944e-09	1.146133e-13	8.297800e-21
Fibcd1	1.303896e-18	1.7561711	0.277	0.088	3.650649e-14	4.053777e-20	1.5887512	0.254	0.065	1.134977e-15	1.303896e-18	8.107554e-20
Col5a2	4.191867e-10	2.1711345	0.110	0.029	1.173639e-05	1.079825e-19	2.5414465	0.107	0.014	3.023293e-15	4.191867e-10	2.159649e-19
Dcn	3.779572e-05	1.1004743	0.141	0.066	1.000000e+00	1.467456e-19	1.2494377	0.243	0.061	4.108582e-15	3.779572e-05	2.934911e-19
Brd9	1.091491e-10	0.5859223	0.806	0.663	3.055958e-06	1.828784e-19	0.8828290	0.870	0.575	5.120229e-15	1.091491e-10	3.657568e-19

# Part 8: Find markers

```
Find markers across conditions
Seurat_AD_WT$cluster_cond <- paste(Seurat_AD_WT$seurat_clusters,
 Seurat_AD_WT$condition, sep = "_")

Idents(Seurat_AD_WT) <- "cluster_cond"
cluster13.markers.cond <- FindMarkers(Seurat_AD_WT, ident.1 = "13_WT", ident.2 = "13_AD")
```

## Condition Markers

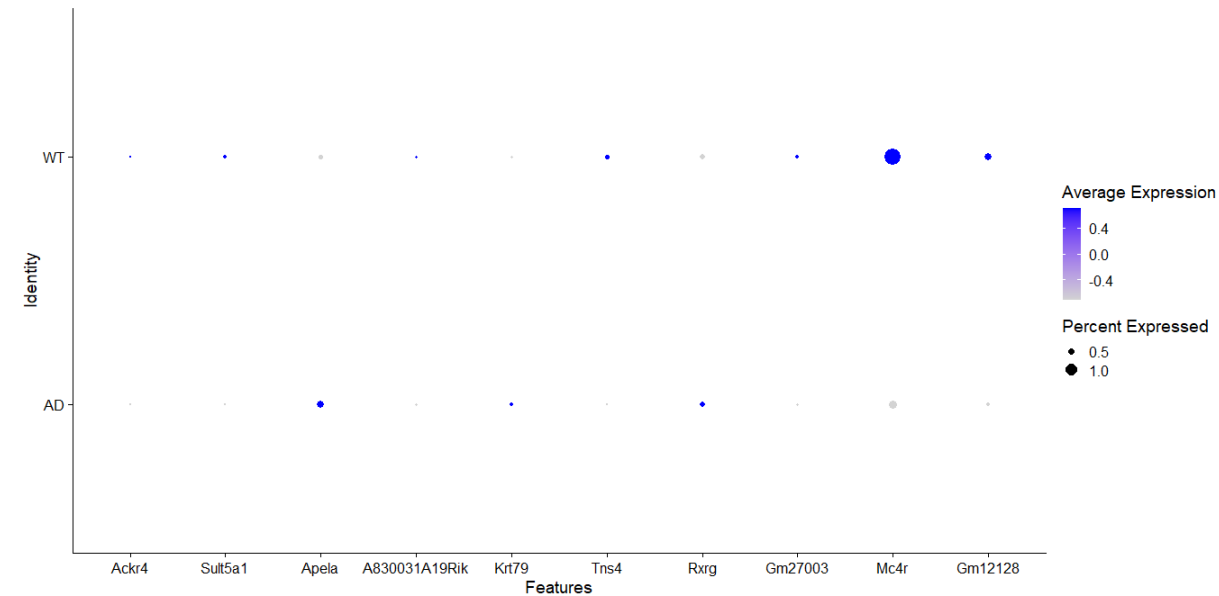
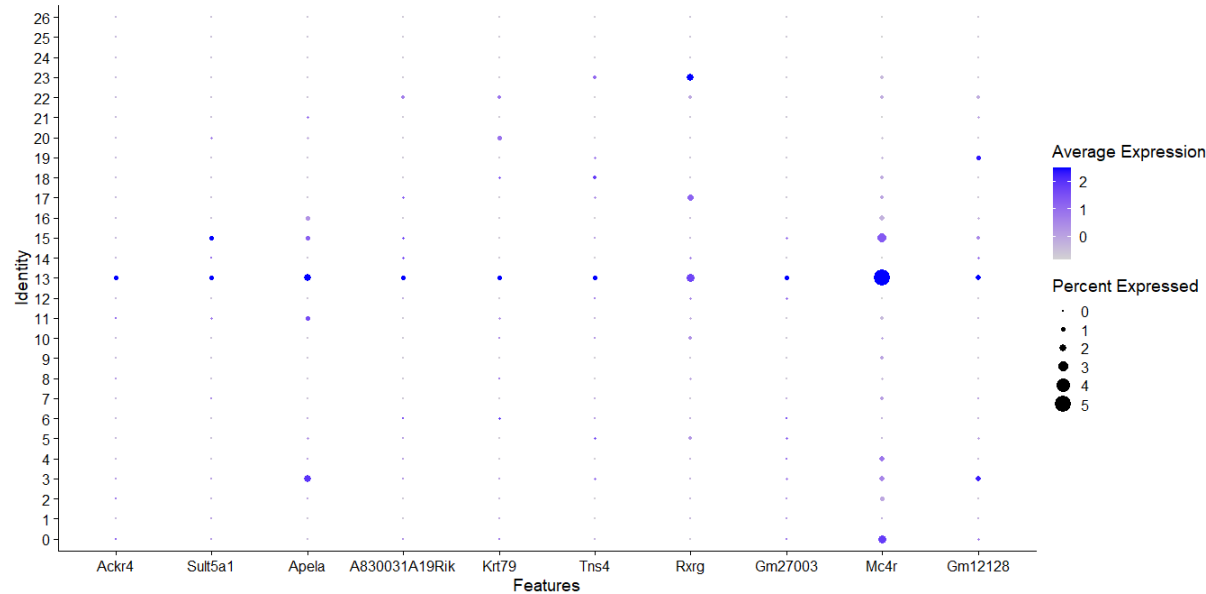


	p_val	avg_log2FC	pct.1	pct.2	p_val_adj
Apoe	5.241195e-46	8.1486988	0.780	0.006	1.467430e-41
Ttr	1.134299e-22	6.7309510	0.455	0.006	3.175811e-18
Rnf121	3.780637e-19	5.6537259	0.424	0.024	1.058503e-14
Gm26917	3.103626e-16	1.1466041	0.880	0.633	8.689531e-12
Gm38039	1.649691e-15	4.2406638	0.356	0.018	4.618805e-11
Eif2s3y	4.668433e-10	1.0712380	0.754	0.479	1.307068e-05
PISD	3.730705e-09	-0.7947013	0.796	0.917	1.044523e-04
Galnt1	1.822043e-06	-0.6015449	0.670	0.846	5.101357e-02
Slc24a5	3.015296e-06	-0.7756558	0.592	0.746	8.442226e-02
Timp4	8.124086e-06	-1.3849407	0.199	0.402	2.274582e-01
Sv2b	8.632651e-06	-0.9854270	0.351	0.550	2.416970e-01
mt-Nd4	8.915913e-06	0.7297622	0.728	0.580	2.496277e-01
AY036118	9.411056e-06	1.9050876	0.251	0.077	2.634907e-01
Ybx1	1.496202e-05	-1.1925602	0.220	0.414	4.189067e-01
Mbp	2.233738e-05	1.2708685	0.393	0.183	6.254018e-01
Ptcra	2.991650e-05	-1.4398573	0.110	0.278	8.376023e-01
Gm14966	4.146767e-05	3.9741931	0.110	0.006	1.000000e+00
Fam160b2	4.180642e-05	2.2633548	0.178	0.041	1.000000e+00
Bcas3	4.746139e-05	0.9550465	0.450	0.237	1.000000e+00

# Part 8: Find markers

```
Plot the markers
`{r, fig.height = 7, fig.width = 14}
cluster13.markers %>%
 dplyr::filter(avg_log2FC > 2 & p_val < 0.05) %>%
 slice_max(avg_log2FC, n = 10) -> top10

Idents(Seurat_AD_WT) <- "seurat_clusters"
DotPlot(Seurat_AD_WT, features = rownames(top10))
DotPlot(Seurat_AD_WT, features = rownames(top10), group.by = "condition")
`}
```



# Part 8: Find markers

Aspect	FindMarkers(ident.1 = 5)	FindMarkers(ident.1 = "5_WT", ident.2 = "5_KO")	FindConservedMarkers(ident.1 = 5, grouping.var = "condition")
Purpose	Find genes that define cluster 5 identity	Find genes differentially expressed between conditions in cluster 5	Find genes that consistently mark cluster 5 regardless of condition
Comparison	Cluster 5 vs. all other cells	WT cluster 5 vs. KO cluster 5	Within-condition cluster-vs-cluster comparisons, then combined
Output significance	Single p-value per gene	Single p-value per gene	Separate p-values per condition + meta-analysis p-value
Biological question	What makes cluster 5 unique?	Which genes respond differently to genotype in cluster 5?	Which genes robustly identify cluster 5 across both genotypes?

# Part 8: Find markers

## Markers

	p_val	avg_log2FC
Cpne7	3.795059e-113	1.6148848
Pex5l	6.549430e-80	1.3841500
Man1a	1.168222e-55	1.7415038
Grin2b	6.088627e-55	0.6456170
Kcnn2	1.985708e-46	1.3211142
3110035E14Rik	1.610460e-42	1.1141428
Ntm	1.879055e-40	0.7167477
Chrd	8.107074e-40	1.0820214
Fam19a1	3.278474e-37	0.6712845
Fibcd1	4.322958e-37	1.6778961
Zeb2	6.237067e-33	0.7754559
Arhgef28	8.172101e-31	1.3445675
Mppcd1	4.221407e-29	1.0897637
Brrl9	7.888922e-28	0.7317513
Cadps2	6.051534e-27	1.3831535
Atp2b1	1.206030e-26	0.5222782
Grn5	2.330329e-26	0.5810710

## Conserved Markers

	WT_p_val	WT_avg_log2FC	WT_pct.1	WT_pct.2	WT_p_val_adj	AD_p_val	AD_avg_log2FC
Cpne7	1.896286e-53	1.5667888	0.921	0.475	5.309222e-49	3.859953e-62	1.6734406
Man1a	4.180960e-17	1.4020101	0.372	0.149	1.170585e-12	7.693521e-47	2.1012876
Pex5l	2.917237e-40	1.4236271	0.859	0.435	8.167680e-36	2.128116e-41	1.3607309
Grin2b	7.679557e-37	0.7202418	1.000	0.929	2.150122e-32	1.357831e-20	0.5502120
Kcnn2	1.643295e-19	1.3106818	0.571	0.288	4.600898e-15	6.643690e-30	1.3249140
Fam19a1	6.108650e-26	0.7810839	0.942	0.578	1.710300e-21	6.415784e-14	0.5597053
Arhgef28	8.345182e-10	1.1449230	0.262	0.114	2.336484e-05	1.180007e-25	1.5509784
Ntm	6.855173e-18	0.6564655	0.953	0.702	1.919311e-13	2.629994e-25	0.7718903
3110035E14Rik	7.409298e-20	0.9887040	0.660	0.348	2.074455e-15	3.045912e-25	1.2567730
Chrd	1.333034e-18	1.0821957	0.660	0.375	3.732227e-14	4.238994e-23	1.0950006
Zeb2	4.148900e-21	0.8652456	0.853	0.613	1.161609e-16	1.146133e-13	0.6508298
Fibcd1	1.303896e-18	1.7561711	0.277	0.088	3.650649e-14	4.053777e-20	1.5887512
Col5a2	4.191867e-10	2.1711345	0.110	0.029	1.173639e-05	1.079825e-19	2.5414465
Dcn	3.779572e-05	1.1004743	0.141	0.066	1.000000e+00	1.467456e-19	1.2494377
Brd9	1.091491e-10	0.5859223	0.806	0.663	3.055958e-06	1.828784e-19	0.8828290

# Thank you for the attention

# Please evaluate our course!

Lukasz

Kaja

# Part 8: Find markers

```
##{r, fig.height = 7, fig.width = 14}
find all markers of cluster 5
Idents(Seurat_AD_WT) <- "seurat_clusters"
cluster13.markers <- FindMarkers(Seurat_AD_WT, ident.1 = "13", only.pos = TRUE)
head(cluster13.markers, n = 10)

Idents(Seurat_AD_WT) <- "seurat_clusters"
cluster13.conservd <- FindConservedMarkers(Seurat_AD_WT, ident.1 = "13", grouping.var = "condition", verbose = FALSE)

Find markers across conditions
Seurat_AD_WT$cluster_cond <- paste(Seurat_AD_WT$seurat_clusters,
 Seurat_AD_WT$condition, sep = "_")

Idents(Seurat_AD_WT) <- "cluster_cond"
cluster13.markers.cond <- FindMarkers(Seurat_AD_WT, ident.1 = "13_WT", ident.2 = "13_AD")
...

Plot the markers
##{r, fig.height = 7, fig.width = 14}
cluster13.markers %>%
 dplyr::filter(avg_log2FC > 2 & p_val < 0.05) %>%
 slice_max(avg_log2FC, n = 10) -> top10

Idents(Seurat_AD_WT) <- "seurat_clusters"
DotPlot(Seurat_AD_WT, features = rownames(top10))
DotPlot(Seurat_AD_WT, features = rownames(top10), group.by = "condition")
...
```

## Markers

	p_val	avg_log2FC	pct.1	pct.2	p_val_adj
Cpne7	3.795059e-113	1.6148848	0.939	0.440	1.062541e-108
Pex5l	6.549430e-80	1.3841500	0.867	0.414	1.833709e-75
Man1a	1.168222e-55	1.7415038	0.408	0.123	3.270787e-51
Grin2b	6.088627e-55	0.6456170	1.000	0.913	1.704694e-50
Kcnn2	1.985708e-46	1.3211142	0.589	0.252	5.559586e-42
3110035E14Rik	1.610460e-42	1.1141428	0.614	0.279	4.508965e-38
Ntm	1.879055e-40	0.7167477	0.956	0.618	5.260977e-36
Chrd	8.107074e-40	1.0820214	0.683	0.354	2.269819e-35
Fam19a1	3.278474e-37	0.6712845	0.917	0.568	9.179072e-33
Fibcd1	4.322958e-37	1.6778961	0.267	0.076	1.210342e-32
Zeb2	6.237067e-33	0.7754559	0.853	0.579	1.746254e-28
Arhgef28	8.172101e-31	1.3445675	0.294	0.099	2.288025e-26
Mppcd1	4.221407e-29	1.0897637	0.481	0.220	1.181910e-24
Brd9	7.888922e-28	0.7317513	0.836	0.619	2.208740e-23
Cadps2	6.051534e-27	1.3831535	0.256	0.086	1.694308e-22
Atp2b1	1.206030e-26	0.5222782	0.939	0.733	3.376643e-22
Grm5	2.330329e-26	0.5810710	0.939	0.721	6.524456e-22

## Conserved Markers

	WT_p_val	WT_avg_log2FC	WT_pct.1	WT_pct.2	WT_p_val_adj	AD_p_val	AD_avg_log2FC	AD_pct.1	AD_pct.2	AD_p_val_adj	max_pval	minimump_p_val
Cpne7	1.896286e-53	1.5667888	0.921	0.475	5.309222e-49	3.859953e-62	1.6734406	0.959	0.406	1.080710e-57	1.896286e-53	7.719906e-62
Man1a	4.180960e-17	1.4020101	0.372	0.149	1.170585e-12	7.693521e-47	2.1012876	0.450	0.097	2.154032e-42	4.180960e-17	1.538704e-46
Pex5l	2.917237e-40	1.4236271	0.859	0.435	8.167680e-36	2.128116e-41	1.3607309	0.876	0.395	5.958298e-37	2.917237e-40	4.256231e-41
Grin2b	7.679557e-37	0.7202418	1.000	0.929	2.150122e-32	1.357831e-20	0.5502120	1.000	0.896	3.801656e-16	1.357831e-20	1.535911e-36
Kcnn2	1.643295e-19	1.3106818	0.571	0.288	4.600898e-15	6.643690e-30	1.3249140	0.609	0.216	1.860100e-25	1.643295e-19	1.328738e-29
Fam19a1	6.108650e-26	0.7810839	0.942	0.578	1.710300e-21	6.415784e-14	0.5597053	0.888	0.557	1.796291e-09	6.415784e-14	1.221730e-25
Arhgef28	8.345182e-10	1.1449230	0.262	0.114	2.336484e-05	1.180007e-25	1.5509784	0.331	0.085	3.303784e-21	8.345182e-10	2.360014e-25
Ntm	6.855173e-18	0.6564655	0.953	0.702	1.919311e-13	2.629994e-25	0.7718903	0.959	0.536	7.363456e-21	6.855173e-18	5.259987e-25
3110035E14Rik	7.409298e-20	0.9887040	0.660	0.348	2.074455e-15	3.045912e-25	1.2567730	0.562	0.212	8.527944e-21	7.409298e-20	6.091824e-25
Chrd	1.333034e-18	1.0821957	0.660	0.375	3.732227e-14	4.238994e-23	1.0950006	0.710	0.334	1.186834e-18	1.333034e-18	8.477988e-23
Zeb2	4.148900e-21	0.8652456	0.853	0.613	1.161609e-16	1.146133e-13	0.6508298	0.852	0.546	3.208944e-09	1.146133e-13	8.297800e-21
Fibcd1	1.303896e-18	1.7561711	0.277	0.088	3.650649e-14	4.053777e-20	1.5887512	0.254	0.065	1.134977e-15	1.303896e-18	8.107554e-20
Col5a2	4.191867e-10	2.1711345	0.110	0.029	1.173639e-05	1.079825e-19	2.5414465	0.107	0.014	3.023293e-15	4.191867e-10	2.159649e-19
Dcn	3.779572e-05	1.1004743	0.141	0.066	1.000000e+00	1.467456e-19	1.2494377	0.243	0.061	4.108582e-15	3.779572e-05	2.934911e-19
Brd9	1.091491e-10	0.5859223	0.806	0.663	3.055958e-06	1.828784e-19	0.8828290	0.870	0.575	5.120229e-15	1.091491e-10	3.657568e-19

## Condition Markers



	p_val	avg_log2FC	pct.1	pct.2	p_val_adj
ApoE	5.241195e-46	8.1486988	0.780	0.006	1.467430e-41
Ttr	1.134299e-22	6.7309510	0.455	0.006	3.175811e-18
Rnf121	3.780637e-19	5.6537259	0.424	0.024	1.058503e-14
Gm26917	3.103626e-16	1.1466041	0.880	0.633	8.689531e-12
Gm38039	1.649691e-15	4.2406638	0.356	0.018	4.618805e-11
Eif2s3y	4.668433e-10	1.0712380	0.754	0.479	1.307068e-05
PLSD	3.730705e-09	-0.7947013	0.796	0.917	1.044523e-04
Galnt1	1.822043e-06	-0.6015449	0.670	0.846	5.101357e-02
Slc24a5	3.015296e-06	-0.7756558	0.592	0.746	8.442226e-02
Timp4	8.124086e-06	-1.3849407	0.199	0.402	2.274582e-01
Sv2b	8.632651e-06	-0.9854270	0.351	0.550	2.416970e-01
mt-Nd4	8.915913e-06	0.7297622	0.728	0.580	2.496277e-01
AY036118	9.411056e-06	1.9050876	0.251	0.077	2.634907e-01
Ybx1	1.496202e-05	-1.1925602	0.220	0.414	4.189067e-01
Mbp	2.233738e-05	1.2708685	0.393	0.183	6.254018e-01
Ptcr	2.991650e-05	-1.4398573	0.110	0.278	8.376023e-01
Gm14966	4.146767e-05	3.9741931	0.110	0.006	1.000000e+00
Fam160b2	4.180642e-05	2.2633548	0.178	0.041	1.000000e+00
Bcas3	4.746139e-05	0.9550465	0.450	0.237	1.000000e+00